LABORATORY MANUAL

CE3102 : Signals, System and Transform

Hardware Lab 1 (Location: N4-01a-03)

SESSION 2024/25

SEMESTER 2

COMPUTER ENGINEERING COURSE

Lab Manual: Discrete Fourier Transform

1. Given *y1(t) = 2cos(2*π*2*t)*, *y2(t) = 3sin(2* π *10*t- π /3)*, and *y3(t) = y1(t)+y2(t)* sampled at *Fs = 100*.
   a. **Sampling of sinusoids**
      i. Find the discrete time representation of *y1[n], y2[n], and y3[n]*.
      ii. Find the period of *y1[n]*, *y2[n]*,and *y3[n]*.
      iii. How are the periods of *y3[n]* related to *y1[1]* and *y2[n]*?

   b. **Generate the DTFS and DFT representation** (magnitude and phase) of *y3[n]*
      i. Using N_0 samples, where N_0= period of *y3[n]*
      ii. Comment on the phase spectrum where there is no signal of interest, is the phase important. How should we deal with the phases when the signal is not present.
      iii. Use $N_0 + 1$ to generate the magnitude spectrum. Comment what is spectral leakage, and why did it occur.
      iv. Use $2N_0$ and $10N_0$ samples, comment on how did it affect the representation along the horizontal axis.

2. Let $W^k = 1e^{j\left(\frac{2\pi}{N}n\right)k}$ and $k \in Z$,
   a. Generate the 2-D polar and 3-D plots of W^k for *N = 50*, *n = 0..(N-1)*, plot $W^{k=0}, W^{k=1}, W^{k=-1}, W^{k=49}$,
   b. Comment on how are $W^{k=1}$ vs $W^{k=-1}$ related ?

3. The impulse response of a LTI system.
   a. State the convolution equation and discuss its application.
   b. What happens to a pure sinusoid when it is passes through an LTI system? What does Eigen function of an LTI system mean?
   c. Generate $x[n] = 1 * \cos(0.1\pi\, n)$ for 5 cycles of the signal. Convolve this signal with $h[n] = \{0.2, 0.3, -0.5\}$, n = 0,1,2. Plot the input and output signal together on the same graph, and comment on their relationship. Calculate the steady state theoretical gain and phase shift to be experiences by x[n] and compare (comment) it to the results obtained via convolution.
   d. redo 3c with h[n]= 0.99.^k, where k = 0:100;